Chapter 3 Exercises and Solution

3.1 The waveform of voltage across a 30μF capacitor is shown in Figure P3-1.

(1) Draw the waveform of current flowing through the capacitor.

(2) Draw the waveforms of power $p(t)$ and energy $w(t)$ for the capacitor.

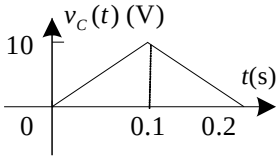
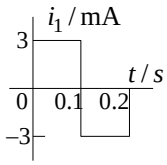


Figure P3-1



(c)

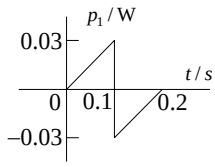
Solution (1)

$$i_1(t) = C\frac{\mathrm{d}v_1(t)}{\mathrm{d}t} = \begin{cases} 0\ \text{A} & -\infty < t < 0 \\ 3\times10^{-3}\ \text{A} & 0 < t < 0.1\text{s} \\ -3\times10^{-3}\ \text{A} & 0.1\text{s} < t < 0.2\text{s} \\ 0\ \text{A} & 0.2\text{s} < t < \infty \end{cases}$$

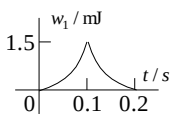
(2)

$$p_1(t) = v_1(t)\times i_1(t) = \begin{cases} 0\ \text{W} & -\infty < t < 0 \\ 0.3t\ \text{W} & 0 < t < 0.1\text{s} \\ 0.3t - 0.06\ \text{W} & 0.1\text{s} < t < 0.2\text{s} \\ 0\ \text{W} & 0.2\text{s} < t < \infty \end{cases}$$

$$w_1(t) = \int_{-\infty}^{t} p_1(\tau)d\tau = \begin{cases} 0\ \text{J} & -\infty < \text{t} < 0 \\ 0.15t^2\ \text{J} & 0 < \text{t} < 0.1\text{s} \\ 0.15t^2 - 0.06t + 0.006\ \text{J} & 0.1\text{s} < \text{t} < 0.2\text{s} \\ 0\ \text{J} & 0.2\text{s} < \text{t} < \infty \end{cases}$$

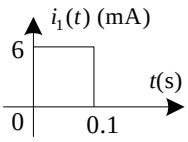


(a)

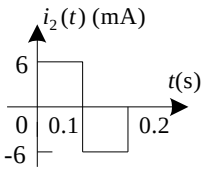


(b)

3.2 The waveforms of the two currents through a 30-μF capacitor are shown in Figure P3-2. Draw the waveforms of corresponding voltages across the capacitor.

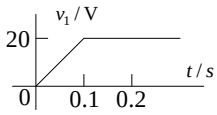


(a)

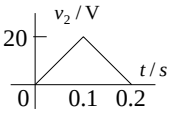


(b)

Figure P3-2



(c)



(d)

Solution (a)

$$v_1(t)=\frac{1}{C}\int_{-\infty}^{t} i_1(\tau)d\tau=\begin{cases}0\ \text{V} & -\infty<t<0\\ 200t\ \text{V} & 0<t<0.1\text{s}\\ 20\ \text{V} & 0.1\text{s}<t<\infty\end{cases}$$

The waveform is shown in Fig.(c).

(b)
$$v_2(t)=\frac{1}{C}\int_{-\infty}^{t} i_2(\tau)d\tau=\begin{cases}0\ \text{V} & -\infty<t<0\\ 200t\ \text{V} & 0<t<0.1\text{s}\\ -200t+40\ \text{V} & 0.1\text{s}<t<0.2\text{s}\\ 0\ \text{V} & 0.2\text{s}<t<\infty\end{cases}$$

The waveform is shown in Fig.(d).

3.3 Draw $v(t)$ waveforms of the following currents through an inductor of L=2mH.

(1) $i(t)=4t+2$ (A)

(2) $i(t)=6\,\mathrm{e}^{-2t}$ (A)

(3) $i(t)=4\cos 10t$ (A)

(4) 6 (A)

Solution

(1) $v=L\dfrac{\mathrm{d}i}{\mathrm{d}t}=2\times10^{-3}\times4=8\times10^{-3}\ \text{V},\quad t>0$

The waveform is shown in Fig.P3-3(a).

(2) $v=L\dfrac{\mathrm{d}i}{\mathrm{d}t}=2\times10^{-3}\times6\times(-2)\mathrm{e}^{-2t}=-24\times10^{-3}\mathrm{e}^{-2t}\ \text{V},\quad t>0$

The waveform is shown in Fig.P3-3(b).

(3) $v=L\dfrac{\mathrm{d}i}{\mathrm{d}t}=2\times10^{-3}\times(-4\times10\sin 10t)=-80\times10^{-3}\sin 10t\ \text{V},\quad t>0$

The waveform is shown in Fig.P3-3(c).

(4) $v=L\dfrac{di}{dt}=2\times10^{-3}\times0=0\ \text{V},\quad t>0$

The waveform is shown in Fig.P3-3(d).

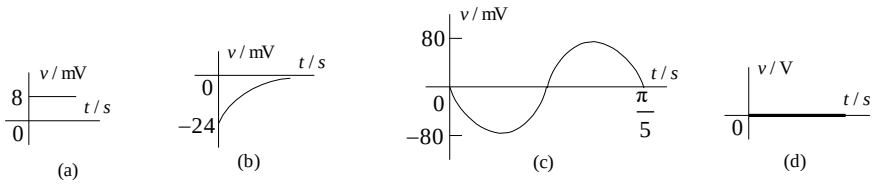


Figure P3-3

3.4 Write the differential equations for the specified variables in Figure P3-4 at $t>0$.

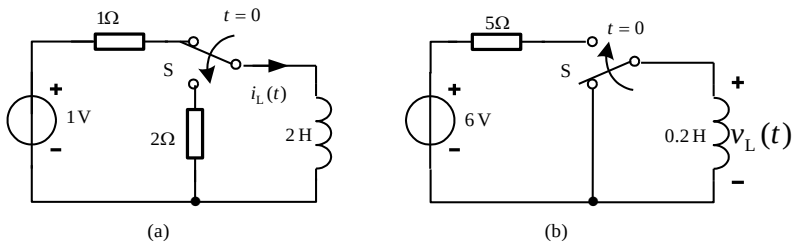


Figure P3-4

Solution

(1) $2\frac{\mathrm{d}i_L}{\mathrm{d}t}+2i_L=0$

$\frac{\mathrm{d}i_L}{\mathrm{d}t}+i_L=0$

(2)

$$\begin{cases}5i_L+v_L=6\\ v_L=0.2\frac{\mathrm{d}i_L}{\mathrm{d}t}\end{cases}$$

$\frac{\mathrm{d}v_L}{\mathrm{d}t}+25v_L=0$

3.5 Write the differential equations for specified variables in Figure P3-5 at $t>0$.

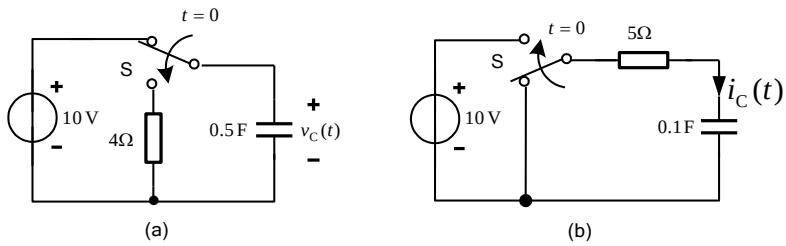


Figure P3-5

Solution

(1) $4\times(0.5\frac{\mathrm{d}v_C}{\mathrm{d}t})+v_C=0$

$\frac{\mathrm{d}v_C}{\mathrm{d}t}+0.5v_C=0$

(2)

$$\begin{cases}5i_C+v_C=10\\ i_C=0.1\frac{\mathrm{d}v_C}{\mathrm{d}t}\end{cases}$$

$\frac{\mathrm{d}i_C}{\mathrm{d}t}+2i_C=0$

3.6 The circuit in Figure P3-6 is working at $t>0$.

(1) Write the differential equations for v and i at $t>0$.

(2) Determine v and i at $t>0$ if given $v_s=3\text{V}$ and $v(0^+)=0$.

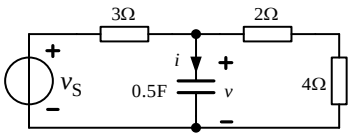


Figure P3-6

Solution

(1)

$$\begin{cases} \dfrac{v-v_s}{3}+0.5\dfrac{\mathrm{d}v}{\mathrm{d}t}+\dfrac{v}{2+4}=0 \\ i=0.5\dfrac{\mathrm{d}v}{\mathrm{d}t} \end{cases}$$

$$\frac{\mathrm{d}v}{\mathrm{d}t}+v=\frac{2}{3}v_s$$

$$\frac{\mathrm{d}i}{\mathrm{d}t}+i=0$$

(2) $v(t)=K\mathrm{e}^{-t}+2=-2\mathrm{e}^{-t}+2\ \mathrm{V}, t>0$

$$i(0^+)=\frac{3}{3}=1\mathrm{A}$$

$$i(t)=K\mathrm{e}^{-t}=\mathrm{e}^{-t}\ \mathrm{A}, t>0$$

3.7 Figure P3-7 shows a circuit at $t > 0$.

(1) Write the differential equation for v.

(2) Find $v(t)$ if $v(0^+)=5\mathrm{V}$.

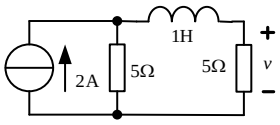


Figure P3-7

Solution

$$(1)\ v+1\times\frac{\mathrm{d}(\frac{v}{5})}{\mathrm{d}t}+5\times(\frac{v}{5}-2)=0$$

$$\frac{\mathrm{d}v}{\mathrm{d}t}+10v=50$$

$$(2)\ v(t)=K\mathrm{e}^{-10t}+5=5\mathrm{V}, t>0$$

3.8 Given $i_1(0^+)=2\,\mathrm{A}$ and $v_\mathrm{C}(0^+)=4\,\mathrm{V}$ in Figure P3-8, find the values of the specified variables at $t=0^+$ and $t=\infty$, respectively.

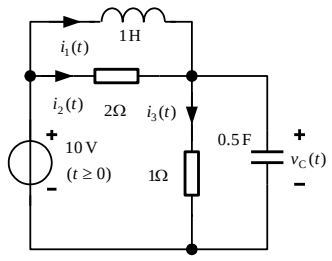
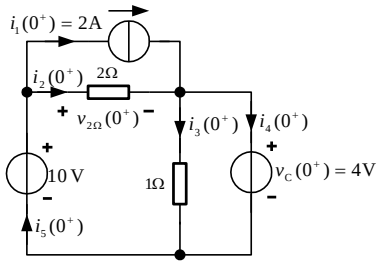


Figure P3-8

Solution The equivalent circuit at t=0⁺ is shown in FigureP3-8(b)，

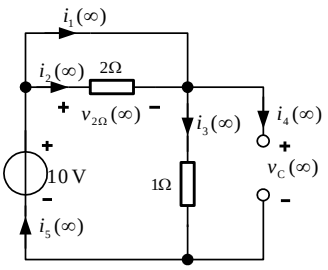


(b)

$$\begin{cases} i_1(0^+) = 2\text{A} \\ i_2(0^+) = \dfrac{10-4}{2} = 3\text{A} \\ i_3(0^+) = \dfrac{4}{1} = 4\text{A} \\ i_4(0^+) = i_1(0^+) + i_2(0^+) - i_3(0^+) = 2+3-4 = 1\text{A} \\ i_5(0^+) = i_1(0^+) + i_2(0^+) = 2+3 = 5\text{A} \end{cases}$$

$$\begin{cases} v_C(0^+) = 4\text{V} \\ v_{2\Omega}(0^+) = 2\times 3 = 6\text{V} \end{cases}$$

The equivalent circuit at $t=\infty$ is shown in FigureP3-8 (c),



(c)

$$\begin{cases} i_1(\infty) = i_3(\infty) = i_5(\infty) = \dfrac{10}{1} = 10\text{A} \\ i_2(\infty) = 0\text{A} \\ i_4(\infty) = 0\text{A} \end{cases}$$

$$\begin{cases} v_C(\infty) = 10\text{V} \\ v_{2\Omega}(\infty) = 0\text{V} \end{cases}$$

3.9 For the circuit in Figure P3-9, find the initial values $i_{L1}(0^+)$, $i_{L2}(0^+)$ and the final values $i_{L1}(\infty)$, $i_{L2}(\infty)$.

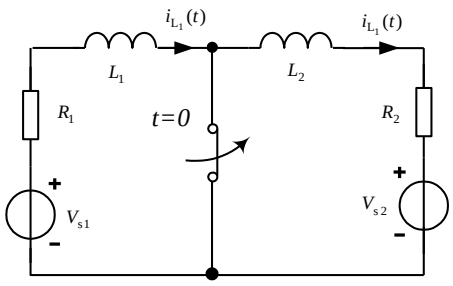


Figure P3-9

Solution

t=0⁻

$$\begin{cases} i_{L_1}(0^-) = \dfrac{V_{s1}}{R_2} \\ i_{L_2}(0^-) = -\dfrac{V_{s2}}{R_2} \end{cases} \quad (1)$$

t=0⁺ (KCL)

$$i_{L_1}(0^+) = i_{L_2}(0^+) \quad (2)$$

In this situation (all inductor branches at the node), according to principle of continuity of magnetic flux $\Psi(0^-)=\Psi(0^+)$

$$L_1 i_{L_1}(0^-) + L_2 i_{L_2}(0^-) = L_1 i_{L_1}(0^+) + L_2 i_{L_2}(0^+) \quad (3)$$

Combine equations（1），（2），（3）, get

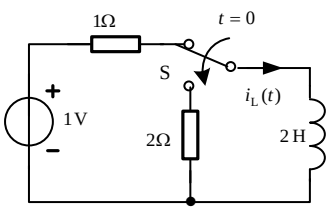
$$i_{L_1}(0^+) = i_{L_2}(0^+) = \frac{1}{L_1 + L_2}\left(\frac{L_1}{R_1}V_{s1} - \frac{L_2}{R_2}V_{s2}\right)$$

At $t=\infty$, set inductors to short circuit, get

$$i_{L_1}(\infty) = i_{L_2}(\infty) = \frac{V_{s1} - V_{s2}}{R_1 + R_2}$$

3.10 Each circuit in Figure P3-4 has been in steady state before the switch acts at t=0. Determine the initial value and solve the differential equation for each specified variable.

Solution



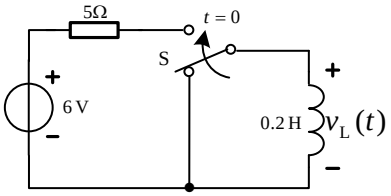
(a)

Figure P3-4

(a) $\dfrac{\mathrm{d}i_L}{\mathrm{d}t} + i_L = 0$

$$i_L(0^+) = 1$$

$$i_L(t) = \mathrm{e}^{-t}\mathrm{A}, t > 0$$



(b)

Figure P3-4

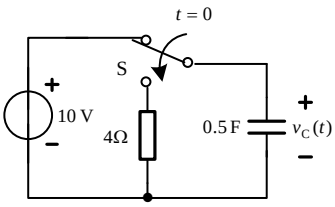
(b) $\dfrac{dv_L}{dt} + 25v_L = 0$

$v_L(0^+) = 6$

$v_L(t) = 6e^{-25t}\text{V}, t > 0$

3.11 Each circuit in Figure P3-5 has been in steady state before the switch acts at t=0. Determine the initial value and solve the differential equation for the specified variable.

Solution



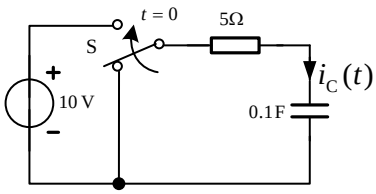
(a)

Figure P3-5

（a） $\dfrac{dv_C}{dt} + 0.5v_C = 0$

$v_C(0^+) = 10\text{V}$

$v_C(t) = 10e^{-0.5t}\text{V}, t > 0$



(b)

Figure P3-5

（b） $\begin{cases} 5i_C + v_C = 10 \\ i_C = 0.1\dfrac{dv_C}{dt} \end{cases}$

$\dfrac{di_C}{dt} + 2i_C = 0$

$i_C(0^+) = 2\text{A}$

$i_C(t) = 2e^{-2t}\text{A}, t > 0$

3.12 If the switch in Figure P3-12 has been open for a long time and is closed at t = 0, find $i_R(0^+)$, $i_L(0^+)$, $i_R(\infty)$ and $i_L(\infty)$, and then write the expressions of i_R(t) and i_L(t).

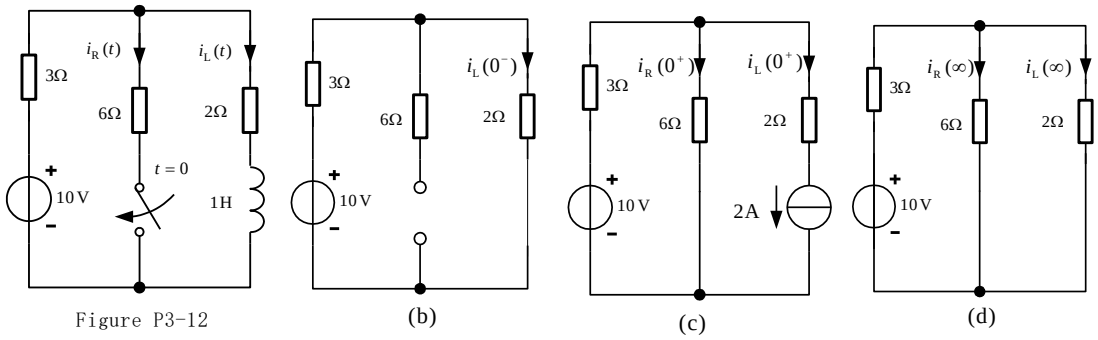


Figure P3-12

Solution

The equivalent circuit at t=0⁻ is shown in FigureP3-12(b),

$$i_L(0^-)=\frac{10}{3+2}=2\text{ A}$$

The equivalent circuit at t=0⁺ is shown in FigureP3-8 (c),

$$i_L(0^+)=i_L(0^-)=2\text{A}$$

$$i_R(0^+)=\frac{10}{3+6}-\frac{3}{3+6}\times 2=\frac{4}{9}\text{A}$$

The equivalent circuit at t=∞is shown in FigureP3-8 (d),

$$i_L(\infty)=\frac{10}{3+6//2}\times\frac{6}{6+2}=\frac{5}{3}\text{A}$$

$$i_R(\infty)=\frac{10}{3+6//2}\times\frac{2}{6+2}=\frac{5}{9}\text{A}$$

The equivalent resistance seen from the inductor is

$$R_0=2+\frac{3\times 6}{3+6}=4\Omega$$

$$\tau=\frac{L}{R_0}=\frac{1}{4}\text{s}$$

$$i_L(t)=[i_L(0^+)-i_L(\infty)]\text{e}^{-\frac{t}{\tau}}+i_L(\infty)=(2-\frac{5}{3})\text{e}^{-4t}+\frac{5}{3}=\frac{5}{3}+\frac{1}{3}\text{e}^{-4t}\text{A},\quad t>0$$

$$i_R(t)=[i_R(0^+)-i_R(\infty)]\text{e}^{-\frac{t}{\tau}}+i_R(\infty)=(\frac{4}{9}-\frac{5}{9})\text{e}^{-4t}+\frac{5}{9}=\frac{5}{9}-\frac{1}{9}\text{e}^{-4t}\text{ A},\quad t>0$$

3.13 The circuit in Figure P3-13 has been in steady state before the switch acts at t=0. Find $i_L(t)$ for $t>0$.

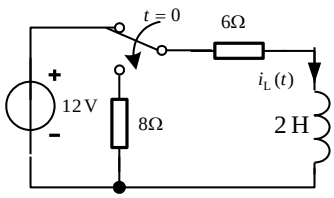


Figure P3-13

Solution

$$i_L(0^-) = \frac{12}{6} = 2\text{A}$$

$$i_L(0^+) = i_L(0^-) = 2\text{A}$$

$$i_L(\infty) = 0\text{A}$$

$$\tau = \frac{2}{8+6} = \frac{1}{7}\text{s}$$

$$i_L(t) = 2\text{e}^{-\frac{t}{1/7}} = 2\text{e}^{-7t}\text{A}, t > 0$$

3.14 The switch in Figure P3-14 has been closed for a long time and is opened at t = 0. Find

(1) $i_1(0^-)$ and $i_2(0^-)$.

(2) $i_1(0^+)$ and $i_2(0^+)$.

(3) $i_1(t)$ for $t > 0$.

(4) $i_2(t)$ for $t > 0$.

(5) Explain why $i_2(0^-) \neq i_2(0^+)$.

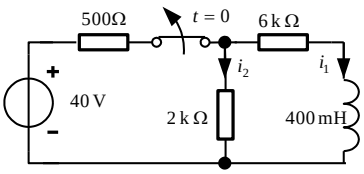
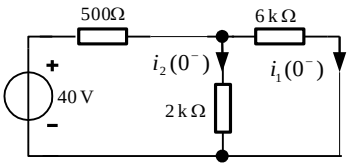
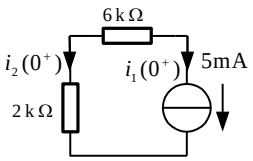


Figure P3-14



(b)



(c)

Solution

（1）The equivalent circuit at t=0$^-$ is shown in FigureP3-14(b)，

$$i_1(0^-) = \frac{40}{500 + 2000 // 6000} \times \frac{2000}{2000 + 6000} = 5\text{mA}$$

$$i_2(0^-) = \frac{40}{500 + 2000 // 6000} \times \frac{6000}{2000 + 6000} = 15\text{mA}$$

(2) The equivalent circuit at t=0$^+$ is shown in FigureP3-14 (c)，

$$i_1(0^+) = 5\text{mA}$$

$$i_2(0^+) = -5\text{mA}$$

(3) $\tau = \dfrac{400 \times 10^{-3}}{2000 + 6000} = 50\,\mu\text{s}$

$$i_1(t) = 5e^{-\frac{t}{50\times10^{-6}}}\text{mA}, \quad t > 0$$

(4) $i_2(t) = -5e^{-\frac{t}{50\times10^{-6}}}\text{mA}, \quad t > 0$

3.15 The circuit at t > 0 is shown in Figure P3-15. Find $v_C(t)$ if $v_C(0^+)$ =0.

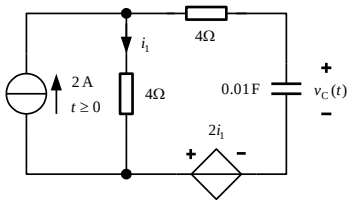
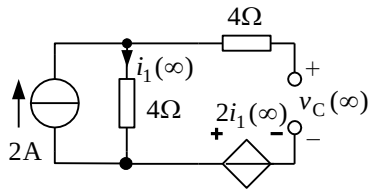
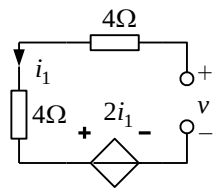


Figure P3-15



(b)



(c)

Solution

The equivalent circuit at $t=\infty$ is shown in FigureP3-15(b),

$v_C(\infty) = 2\times 4 + 2\times 2 = 12\text{V}$

The equivalent circuit for Thevenin resistance is shown in FigureP3-15(c),

$$v = 4i_1 + 4i_1 + 2i_1$$

$$R_0 = \frac{v}{i_1} = 10\Omega$$

$$\tau = 10\times 0.01 = 0.1\text{s}$$

$$v_C(t) = -12\text{e}^{-10t} + 12\text{V}, t > 0$$

3.16 The circuit in Figure P3-16 has been in steady state before the switch acts. Find $v(t)$ and $i(t)$ at $t > 0$.

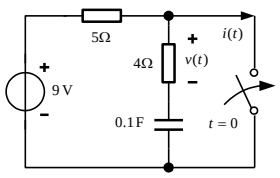
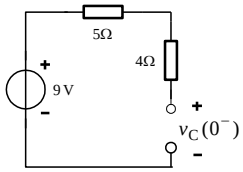
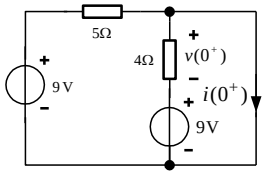


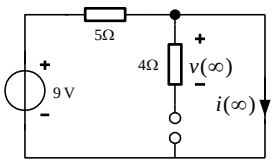
Figure P3-16



(b)



(c)



(d)

Solution

The equivalent circuit at $t=0^-$ is shown in FigureP3-16(b),

$$v_C(0^-) = 9\text{V}$$

The equivalent circuit at $t=0^+$ is shown in FigureP3-16(c),

$$v(0^+) = -9\text{V}$$

$$i(0^+) = \frac{9}{5} + \frac{9}{4} = \frac{81}{20}\text{A}$$

The equivalent circuit at $t=\infty$ is shown in FigureP3-16(d),

$$v(\infty) = 0$$

$$i(\infty) = \frac{9}{5}\text{A}$$

$$\tau = 4\times 0.1 = 0.4\text{s}$$

$$v(t) = -9\text{e}^{-\frac{5}{2}t}\ \text{V}, \quad \text{t} > 0$$

$$i(t) = (\frac{81}{20} - \frac{9}{5})\text{e}^{-\frac{5}{2}t} + \frac{9}{5} = \frac{9}{4}\text{e}^{-\frac{5}{2}t} + \frac{9}{5}\text{A}, t > 0$$

3.17 The switch in Figure P3-17 has been in position 1 for a long time and turns to position 2 at t =0. Find $i_0(t)$ at $t > 0$.

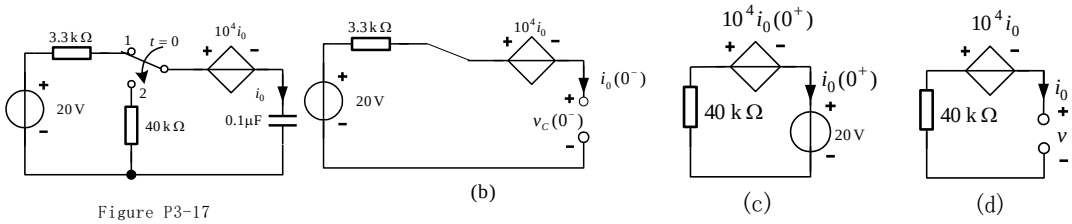


Figure P3-17

Solution

The equivalent circuit at t=0⁻is shown in FigureP3-17(b),

$$v_C(0^-) = 20\,\text{V}$$

$$v_C(0^+) = v_C(0^-) = 20\,\text{V}$$

The equivalent circuit at t=0⁺is shown in FigureP3-17(c),

$$40000\ i_0(0^+) + 10000\ i_0(0^+) + 20 = 0$$

$$i_0(0^+) = -4\times10^{-4}\,\text{A}$$

$$i_0(\infty) = 0$$

The equivalent circuit for Thevenin resistance is shown in FigureP3-17(d),

$$40000\ i_0 + 10000\ i_0 + v = 0$$

$$v = -50000\ i_0$$

$$R_0 = 50000\ \Omega$$

$$\tau = 50000\ \times(0.1\times10^{-6}) = 5\times10^{-3}\,\text{s}$$

$$i_0(t) = -4\times10^{-4}\times e^{-200t}\,\text{A}, t > 0$$

3.18 For the circuit in Figure P3-18, if $i_L(t) = -2e^{-20t} + 1$ A for $t > 0$, find $i_L(0)$, R and L.

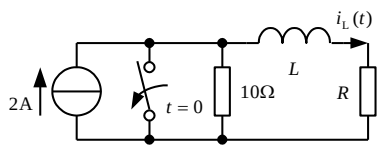


Figure P3-18

Solution

$$i_L(0^+) = -1\text{A}$$

$$i_L(\infty) = 1 = \frac{10}{10+R}\times 2 \Rightarrow R = 10\Omega$$

$$\tau = \frac{1}{20} = \frac{L}{10+10} \Rightarrow L = 1\text{H}$$

3.19 The circuit in Figure P2-19 is in steady state when node a and b are suddenly short-circuited at t=0. For the current i_{ab} flowing through short-circuit wire, find

(1) Initial value i_{ab} (0^+).

(2) Final value $i_{ab}(\infty)$.

(3) Time needed for i_{ab} to rise to 114A.

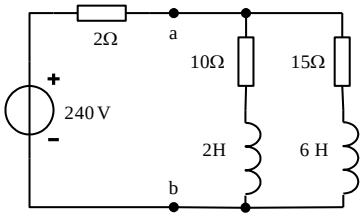
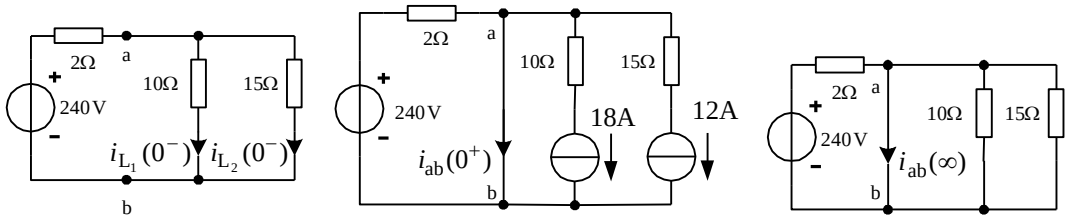


Figure P3-19



(b) (c) (d)

Solution

（1）The equivalent circuit at t=0$^-$is shown in FigureP3-19(b)，

$$i_{L_1}(0^-)=\frac{240}{2+10//15}\times\frac{15}{10+15}=18\text{A}$$

$$i_{L_2}(0^-)=\frac{240}{2+10//15}\times\frac{10}{10+15}=12\text{A}$$

The equivalent circuit at t=0$^+$is shown in FigureP3-19(c)，

$$i_{ab}(0^+)=\frac{240}{2}-18-12=90\text{A}$$

（2）the steady-state equivalent circuit is shown in Figure P3-26(d)，

$$i_{ab}(\infty)=120\text{ A}$$

（3） $i_{ab}(t)=\frac{240}{2}-i_{L_1}(t)-i_{L_2}(t)=120-18\text{e}^{-\frac{t}{2/10}}-12\text{e}^{-\frac{t}{6/15}}=120-18\text{e}^{-5t}-12\text{e}^{-2.5t}$ A

$$114=120-18\text{e}^{-5t_0}-12e^{-2.5t_0}$$

$$t_0=\frac{2}{5}\ln 3\text{ s}$$

3.20 In Figure P3-20, if given $v_s=1\text{V}$, the response is $v_C(t)=1.5e^{-at}+0.5\text{ V}$ at $t>0$. Analyze the same circuit under the same state and find i_C(t) at $t>0$ if given v_s =2V.

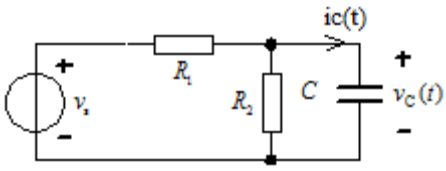


Figure P3-20

Solution

$$v_C=(2-0.5)\text{e}^{-at}+0.5$$

When $v_s = 2\text{V}$, get

$$v_C = (2 - 2\times 0.5)\text{e}^{-at} + 2\times 0.5$$
$$= \text{e}^{-at} + 1\text{V},\ t > 0$$

3.21 In Figure P3-21, the response v_C is $v_C(t) = 2e^{-t} - 3e^{-2t} + 1\text{ V}$ at $t > 0$. If the source changes to v_s=2V, v_c will become $v_C(t) = -2e^{-2t} + 2\text{ V}$ at $t > 0$. Determine zero-input, zero-state, the transient and steady state responses of v_c when v_s=1V.

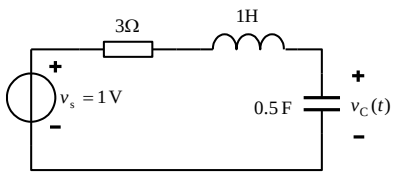


Figure P3-21

Solution

When $v_s = 1\text{V}$

$$v_{C1}(t) = v_{Cx}(t) + v_{Cf}(t) = 2\text{e}^{-t} - 3\text{e}^{-2t} + 1 \qquad \textbf{(1)}$$

When $v_s = 2\text{V}$

$$v_{C2}(t) = v_{Cx}(t) + 2v_{Cf}(t) = -2\text{e}^{-2t} + 2 \qquad \textbf{(2)}$$

Let equation (2) – equation (1), get

$$v_{CX}(t) = (4\text{e}^{-t} - 4\text{e}^{-2t})\text{V}$$

$$v_{Cf}(t) = (-2\text{e}^{-t} + \text{e}^{-2t} + 1)\text{V}$$

$$v_{Ct}(t) = v_{Ch}(t) = 2e^{-t} - 3e^{-2t}\ \text{V}$$

$$v_{Cs}(t) = v_{Cp}(t) = 1\text{ V}$$

3.22 The switch in Fgure P3-22 has been in position *a* for a long time and turns to position *b* at t = 0. At $t > 0$:

(1) Find zero-input and zero-state responses of $i_0(t)$ and $v_0(t)$.

(2) Find transient and steady state responses of $i_0(t)$ and $v_0(t)$.

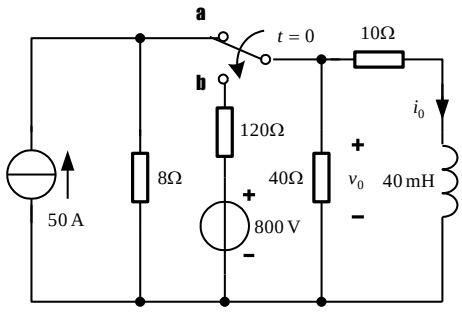
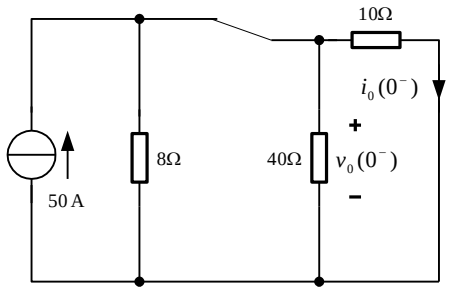
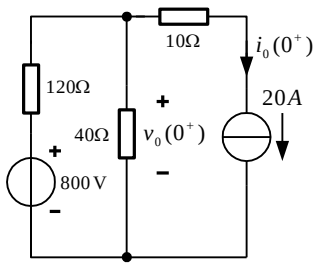


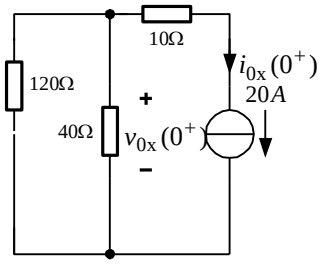
Figure P3-22



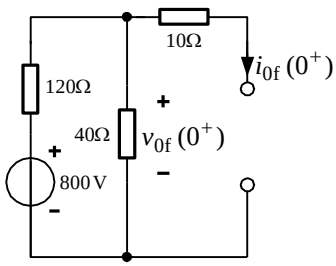
(b)



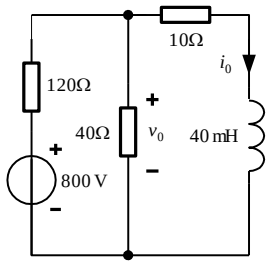
(c)



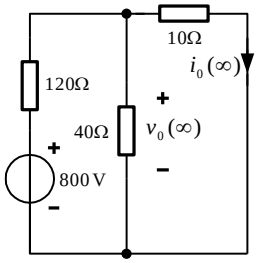
(d)



(e)



(f)



(g)

Solution 1

List differential equation

The equivalent circuit at t=0$^-$is shown in FigureP3-22(b),

$$i_0(0^-)=\frac{\frac{1}{10}}{\frac{1}{8}+\frac{1}{40}+\frac{1}{10}}\times 50=20\,\mathrm{A}$$

The equivalent circuit at t=0$^+$is shown in FigureP3-22(c),

$$i_0(0^+)=20\,\mathrm{A}$$

$$v_0(0^+)=\frac{40}{120+40}\times 800-\frac{120\times 40}{120+40}\times 20=-400\,\mathrm{V}$$

The equivalent circuit for zero-input at t=0$^+$is shown in FigureP3-22(d),

$$\begin{cases} i_{0x}(0^+)=20\,\mathrm{A} \\ v_{0x}(0^+)=-\dfrac{120\times 40}{120+40}\times 20=-600\,\mathrm{V} \end{cases}$$

The equivalent circuit for zero-state at t=0$^+$is shown in FigureP3-22(e),

$$\begin{cases} i_{0f}(0^+) = 0\text{A} \\ v_{0f}(0^+) = \dfrac{40}{120+40} \times 800 = 200\text{ V} \end{cases}$$

The circuit after switch action is shown in FigureP3-22(f),

List differential equations for v_0 and i_0,

$$\begin{cases} v_0 + (\dfrac{v_0}{40} + i_0) \times 120 = 800 \\ -v_0 + 10 i_0 + 40 \times 10^{-3} \dfrac{\mathrm{d}i_0}{\mathrm{d}t} = 0 \end{cases}$$

get

$$\begin{cases} \dfrac{\mathrm{d}i_0}{\mathrm{d}t} + 1000\, i_0 = 5000 \\ \dfrac{\mathrm{d}v_0}{\mathrm{d}t} + 1000\, v_0 = 50000 \end{cases}$$

The differential equations for zero-input are

$$\begin{cases} \dfrac{\mathrm{d}i_{0x}}{\mathrm{d}t} + 1000\, i_{0x} = 0 \\ i_{0x}(0^+) = 20\text{ A} \end{cases} \quad \text{or} \quad \begin{cases} \dfrac{\mathrm{d}v_{0x}}{\mathrm{d}t} + 1000\, v_{0x} = 0 \\ v_{0x}(0^+) = -600\text{ V} \end{cases}$$

Then get the zero-input responses

$$\begin{cases} i_{0x}(t) = 20\,\mathrm{e}^{-1000t}\text{A}, t > 0 \\ v_{0x}(t) = -600\,\mathrm{e}^{-1000t}\text{V}, t > 0 \end{cases}$$

The differential equations for zero-state are

$$\begin{cases} \dfrac{\mathrm{d}i_{0f}}{\mathrm{d}t} + 1000\, i_{0f} = 5000 \\ i_{0f}(0^+) = 0\text{A} \end{cases} \quad \text{or} \quad \begin{cases} \dfrac{\mathrm{d}v_{0f}}{\mathrm{d}t} + 1000\, v_{0f} = 50000 \\ v_{0f}(0^+) = 200\text{ V} \end{cases}$$

Get the zero-state responses

$$\begin{cases} i_{0f}(t) = 5(1 - \mathrm{e}^{-1000t})\text{A}, t > 0 \\ v_{0f}(t) = 50 + 150\,\mathrm{e}^{-1000t}\text{V}, t > 0 \end{cases}$$

The complete responses are

$$\begin{cases} i_0(t) = i_{0x}(t) + i_{0f}(t) = 5 + 15\,\mathrm{e}^{-1000t}\text{A}, t > 0 \\ v_0(t) = v_{0x}(t) + v_{0f}(t) = 50 - 450\,\mathrm{e}^{-1000t}\text{V}, t > 0 \end{cases}$$

Transient and steady-state responses are

$$\begin{cases} i_{0t}(t) = 15 e^{-1000t}\text{A} \\ i_{0s}(t) = 5\text{A} \end{cases} \text{和} \begin{cases} v_{0t}(t) = -450\,\mathrm{e}^{-1000t}\text{V} \\ v_{0s}(t) = 50\text{ V} \end{cases}$$

Solution 2:（three critical factors）

The equivalent circuit for steady-state at $t=\infty$ is shown in FigureP3-22(g),

$$\begin{cases} i_0(\infty) = \dfrac{800}{120 + 40 // 10} \times \dfrac{40}{40 + 10} = 5\text{A} \\ v_0(\infty) = \dfrac{40 // 10}{120 + 40 // 10} \times 800 = 50\,\text{V} \end{cases}$$

$$\begin{cases} i_{0\text{x}}(\infty) = 0\text{A} \\ v_{0\text{x}}(\infty) = 0\text{V} \end{cases}$$

$$\begin{cases} i_{0\text{f}}(\infty) = 5\text{A} \\ v_{0\text{f}}(\infty) = 50\,\text{V} \end{cases}$$

$$\tau = \frac{40 \times 10^{-3}}{120 // 40 + 10} = 10^{-3}\text{s}$$

From solution 1 get

$$\begin{cases} i_{0\text{x}}(0^+) = 20\,\text{A} \\ v_{0\text{x}}(0^+) = -600\,\text{V} \end{cases} \quad \text{and} \quad \begin{cases} i_{0\text{f}}(0^+) = 0\text{A} \\ v_{0\text{f}}(0^+) = 200\,\text{V} \end{cases}$$

Applying them into the short-cut formula, get

$$\begin{cases} i_{0\text{x}}(t) = i_{0\text{x}}(\infty) + [i_{0\text{x}}(0^+) - i_{0\text{x}}(\infty)]\text{e}^{-\frac{t}{\tau}} = 20\text{e}^{-1000t}\text{A}, t > 0 \\ v_{0\text{x}}(t) = v_{0\text{x}}(\infty) + [v_{0\text{x}}(0^+) - v_{0\text{x}}(\infty)]\text{e}^{-\frac{t}{\tau}} = -600\,\text{e}^{-1000t}\text{V}, t > 0 \end{cases}$$

$$\begin{cases} i_{0\text{f}}(t) = i_{0\text{f}}(\infty) + [i_{0\text{f}}(0^+) - i_{0\text{f}}(\infty)]\text{e}^{-\frac{t}{\tau}} = 5(1 - \text{e}^{-1000t})\text{A}, t > 0 \\ v_{0\text{f}}(t) = v_{0\text{f}}(\infty) + [v_{0\text{f}}(0^+) - v_{0\text{f}}(\infty)]\text{e}^{-\frac{t}{\tau}} = 50 + 150\,\text{e}^{-1000t}\text{V}, t > 0 \end{cases}$$

Complete response are

$$\begin{cases} i_0(t) = i_{0\text{x}}(t) + i_{0\text{f}}(t) = 5 + 15\,\text{e}^{-1000t}\text{A}, t > 0 \\ v_0(t) = v_{0\text{x}}(t) + v_{0\text{f}}(t) = 50 - 450\,\text{e}^{-1000t}\text{V}, t > 0 \end{cases}$$

Transient and steady-state responses are

$$\begin{cases} i_{0\text{t}}(t) = 15\,\text{e}^{-1000t}\text{A}, t > 0 \\ i_{0\text{s}}(t) = 5\text{A}, t > 0 \end{cases} \quad \text{and} \quad \begin{cases} v_{0\text{t}}(t) = -450\,\text{e}^{-1000t}\text{V}, t > 0 \\ v_{0\text{s}}(t) = 50\,\text{V}, t > 0 \end{cases}$$

3.23 Express the following signals in terms of step functions .

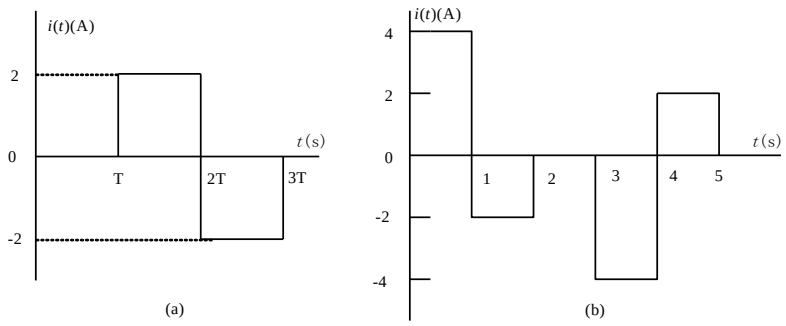


Figure P3-23

Solution

（a） $i(t) = 2u(t-T) - 4u(t-2T) + 2u(t-3T)$

（b） $i(t) = 4u(t) - 6u(t-1) + 2u(t-2) - 4u(t-3) + 6u(t-4) - 2u(t-5)$

3.24 The voltage $v_s(t)$ in the circuit in Figure P3-24(b) is a pulse signal as shown in Figure P3-24(a). Find $i_L(t)$.

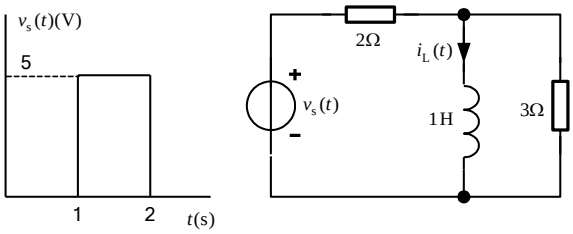


Figure P3-24

Solution 1：（superposition method）

$v_s(t) = 5u(t-1) - 5u(t-2)$

When $5u(t-1)$ works alone, get

$$\begin{cases} i_L(1^+) = 0 \\ i_L(\infty) = \dfrac{5}{2} = 2.5\text{A} \\ \tau = \dfrac{1}{2//3} = \dfrac{5}{6}\text{s} \end{cases}$$

$$i_L(t) = i_L(\infty) + [i_L(1^+) - i_L(\infty)]\mathrm{e}^{-\frac{t-1}{\tau}} = 2.5(1-\mathrm{e}^{-1.2(t-1)})u(t-1)\text{A}$$

When $-5u(t-1)$ work alone, get

$$\begin{cases} i_L(2^+) = 0 \\ i_L(\infty) = -\dfrac{5}{2} = -2.5\text{A} \\ \tau = \dfrac{1}{2//3} = \dfrac{5}{6}\text{s} \end{cases}$$

$$i_L(t) = i_L(\infty) + [i_L(2^+) - i_L(\infty)]\mathrm{e}^{-\frac{t-2}{\tau}} = -2.5(1-\mathrm{e}^{-1.2(t-2)})u(t-2)\text{A}$$

According to superposition principle, get

$$i_L(t) = 2.5(1-\mathrm{e}^{-1.2(t-1)})u(t-1) - 2.5(1-\mathrm{e}^{-1.2(t-2)})u(t-2)\text{A}$$

Solution 2：（piecewise method）

$1\text{s} < t < 2\text{s}$

$$\begin{cases} i_L(1^+) = 0 \\ i_L(\infty) = \dfrac{5}{2} = 2.5\text{A} \\ \tau = \dfrac{1}{2//3} = \dfrac{5}{6}\text{s} \end{cases}$$

$$i_L(t) = i_L(\infty) + [i_L(1^+) - i_L(\infty)]e^{-\frac{t-1}{\tau}} = 2.5(1 - e^{-1.2(t-1)})\text{A}$$

$t > 2\text{s}$

$$\begin{cases} i_L(2^+) = i_L(2^-) = 2.5(1 - e^{-1.2}) \\ i_L(\infty) = 0 \\ \tau = \dfrac{1}{2 // 3} = \dfrac{5}{6}\text{s} \end{cases}$$

Applying them into the short-cut formula, get

$$i_L(t) = i_L(\infty) + [i_L(2^+) - i_L(\infty)]e^{-\frac{t-2}{\tau}} = 2.5(1 - e^{-1.2})e^{-1.2(t-2)}\text{A}$$

3.25 For the circuit in Figure P3-25(a),

(1) find the step response of i(t).

(2) find the zero-state response of i(t) when the circuit is excited by v_s shown in Figure P3-25(b).

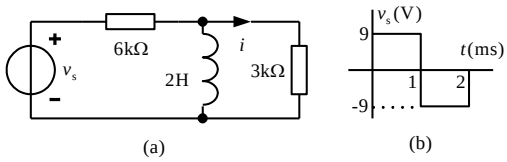


Figure P3-25

（1）When $u(t)$ works alone

$$\begin{cases} i(0^+) = \dfrac{1}{6+3} = \dfrac{1}{9}\text{mA} \\ i(\infty) = 0\text{A} \\ \tau = \dfrac{2}{6 // 3} = 1\text{ms} \end{cases}$$

$$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-\frac{t}{\tau}} = \frac{1}{9}e^{-1000t}u(t)\text{mA}$$

（2） $v_s(t) = 9u(t) - 18u(t - 0.001) + 9u(t - 0.002)$

According to superposition principle, get

$$i(t) = e^{-1000t}u(t) - 2e^{-1000(t-0.001)}u(t - 0.001) + e^{-1000(t-0.002)}u(t - 0.002)\text{mA}$$

3.26 In Figure P3-26, when the voltage across the light rises to 15V, the light turns on and is equivalent to a 10-kΩ resistor. Once the light is on, it will turn off only when the voltage across the light falls below 5V. When it turns off, it is equivalent to an open circuit. Assume the circuit has been working for a long time, and suppose the light turns off at t = 0.

(1) Find the expression of voltage across the light for one cycle.

(2) Determine how many times the light turns on in one minute.

(3) Use Multisim simulation software to find a resistor to replace the 800-kΩ resistor so that the light turns on exactly 12 times per minute.

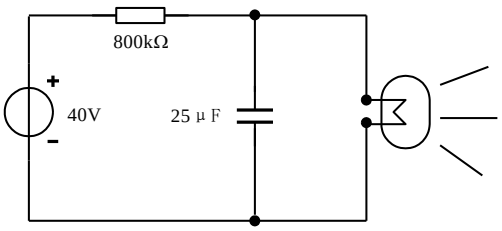
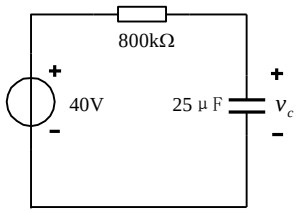
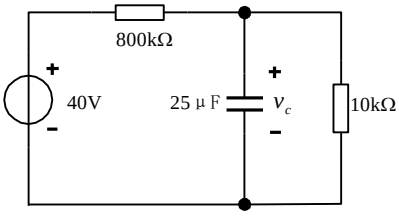


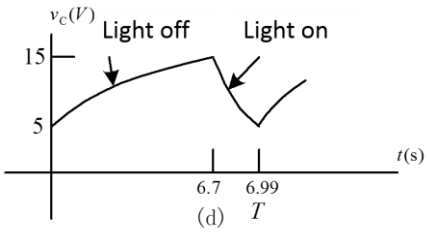
Figure P3-26



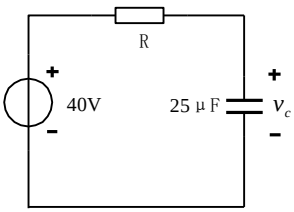
(b)



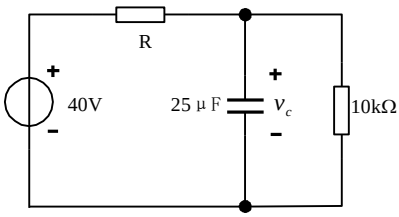
(c)



(d)



(e)



(f)

Solution

（1）In the first period

[light off] $0 < t < t_1$, the equivalent circuit is shown in Figure P-26(b),

$$\begin{cases} v_C(0^+) = 5\text{V} \\ v_C(\infty) = 40\text{V} \\ \tau_1 = 800 \times 10^3 \times 25 \times 10^{-6} = 20\text{s} \end{cases}$$

$$v_C(t) = v_C(\infty) + [v_C(0^+) - v_C(\infty)]\text{e}^{-\frac{t}{\tau_1}} = 40 - 35\text{e}^{-\frac{t}{20}}\text{V}$$

At t_1, the light turns on, get

$$15 = 40 - 35\text{e}^{-\frac{t_1}{20}}$$

$$t_1 = -20\ln\frac{40-15}{35} \approx 6.7s$$

[light on] $t_1 < t < t_2$, the equivalent circuit is shown in Figure P-26 (c),

$$\begin{cases} v_C(t_1^+) = v_C(t_1^-) = 15\text{V} \\ v_C(\infty) = \dfrac{10}{10+800} \times 40 \approx 0.5\text{V} \\ \tau_2 = (800 // 10) \times 10^3 \times 25 \times 10^{-6} \approx 0.25\text{s} \end{cases}$$

$$v_C(t) = v_C(\infty) + [v_C(t_1^+) - v_C(\infty)]\text{e}^{-\frac{t-t_1}{\tau_2}} = 0.5 + 14.5\text{e}^{-\frac{t-6.7}{0.25}}\text{V}$$

At t_2, the light turns off again,

$$5 = 0.5 + 14.5\mathrm{e}^{-\frac{t_2 - 6.7}{0.25}}$$

$$t_2 = -0.25 \ln \frac{5 - 0.5}{14.5} + 6.7 = 0.29 + 6.7 = 6.99\mathrm{s}$$

The voltage waveform on light is shown in Figure P3-26(d).

（2）The period from light off to light on is T=6.99s， in one minute, we have

$$60 = 8 \times T + 4.08$$

4.08s < 6.7s

It shows that the light is turned on 8 times in one minute.

（3）In the range $0<t<t_1$, the equivalent circuit is shown in Figure P-26 (e),

$$\begin{cases} v_{\mathrm{C}}(0^+) = 5\mathrm{V} \\ v_{\mathrm{C}}(\infty) = 40\,\mathrm{V} \\ \tau_1 = R \times 25 \times 10^{-6}\mathrm{s} \end{cases}$$

$$v_{\mathrm{C}}(t) = v_{\mathrm{C}}(\infty) + [v_{\mathrm{C}}(0^+) - v_{\mathrm{C}}(\infty)]\mathrm{e}^{-\frac{t}{\tau_1}} = 40 - 35\mathrm{e}^{-\frac{t}{\tau_1}}\,\mathrm{V}$$

At t_1, the light turns on, get

$$15 = 40 - 35\mathrm{e}^{-\frac{t_1}{\tau_1}}$$

$$t_1 = -\tau_1 \ln \frac{40 - 15}{35} \approx 0.34\tau_1 = 0.34 \times R \times 25 \times 10^{-6} = 8.5R \times 10^{-6}\mathrm{s}$$

In the range $t_1<t<t_2$ (light on), the equivalent circuit is shown in Figure P-26(f),

$$\begin{cases} v_{\mathrm{C}}(t_1^+) = v_{\mathrm{C}}(t_1^-) = 15\,\mathrm{V} \\ v_{\mathrm{C}}(\infty) = \dfrac{10}{10 + R} \times 40 = \dfrac{400}{10 + R} \\ \tau_2 = (R // 10) \times 10^3 \times 25 \times 10^{-6} \end{cases}$$

$$v_{\mathrm{C}}(t) = v_{\mathrm{C}}(\infty) + [v_{\mathrm{C}}(t_1^+) - v_{\mathrm{C}}(\infty)]\mathrm{e}^{-\frac{t - t_1}{\tau_2}} = v_{\mathrm{C}}(\infty) + [15 - v_{\mathrm{C}}(\infty)]\mathrm{e}^{-\frac{t - t_1}{\tau_2}}$$

At t_2, the light turns off, get

$$5 = v_{\mathrm{C}}(\infty) + [15 - v_{\mathrm{C}}(\infty)]\mathrm{e}^{-\frac{t_2 - t_1}{\tau_2}}$$

$$t_2 - t_1 = -\tau_2 \ln \frac{5 - v_{\mathrm{C}}(\infty)}{15 - v_{\mathrm{C}}(\infty)}$$

t_2-t_1 is the time while light is on and it will reach its maximum when R=∞. The time constant will be

$$\tau_2 = 10 \times 10^3 \times 25 \times 10^{-6} = 0.25\mathrm{s}$$

So the longest time for light on is

$$t_{\text{light on}} = -(10 \times 10^3) \times (25 \times 10^{-6}) \times \ln \frac{5}{15} = 0.27\ \mathrm{s}$$

That means the time for light on in each period must be 0~0.27s. If the light needs to be on 12 times a minute, the period will be 5s. to estimate R, neglect the time for light being on， get

$$60 = 12 \times 8.5R \times 10^{-6}$$
$$R = 588\,\text{k}\Omega$$

Since the resistance is more than 500k Ω which is 50 times the resistance while light is on, we can calculate approximately as following.

[light off] $0<t<t_1$, same as before, we have

$$t_1 = -\tau_1 \ln \frac{40-15}{35} \approx 0.34\tau_1 \text{s}$$

[light on] $t_1<t<t_2$ (before light off next time), the equivalent circuit is shown in Figure P-26(f),

$$\begin{cases} v_C(t_1^+) = v_C(t_1^-) = 15\text{V} \\ v_C(\infty) = \dfrac{10}{10+R} \times 40 = \dfrac{400}{10+R} \approx 0 \\ \tau_2 = (R//10) \times 10^3 \times 25 \times 10^{-6} \approx 10 \times 10^3 \times 25 \times 10^{-6} = 0.25\text{s} \end{cases}$$

$$v_C(t) = v_C(\infty) + [v_C(t_1^+) - v_C(\infty)]\text{e}^{-\frac{t-t_1}{\tau_2}} = 15e^{-\frac{t-t_2}{0.25}}\text{V}$$

At t_2 while light is off,

$$5 = 15\text{e}^{-\frac{t_2-t_2}{0.25}}$$
$$t_2 = -0.25 \ln \frac{5}{15} + t_1$$
$$= 0.27 + 0.34 \times 25R \times 10^{-6}$$
$$= 0.27 + 8.5R \times 10^{-6}$$

So

$$60 = 12 \times (0.27 + 8.5R \times 10^{-6})$$
$$R = 556\,\text{k}\Omega$$

3.27 For the circuits in Figure P3-27,

(1) write the differential equations for variables $v_C(t)$ and $i_L(t)$.

(2) solve the equations and find $v_C(t)$ and $i_L(t)$ if given $v_C(0^+)$=5V and $i_L(0^+)$=0.

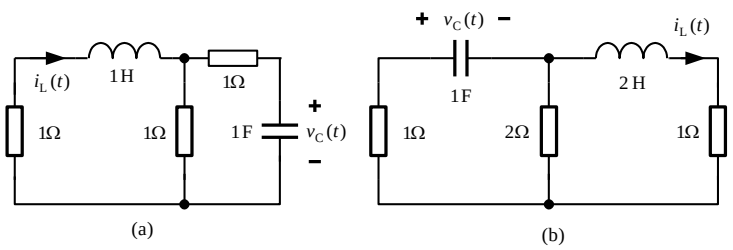
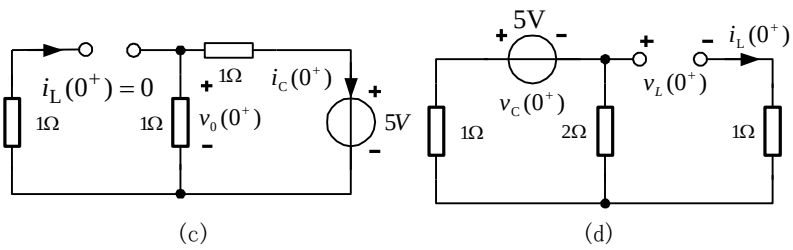


Figure P3-27



Solution

(a) List differential equations

$$\begin{cases} 1\times i_{\mathrm{L}}+1\times\dfrac{\mathrm{d}i_{\mathrm{L}}}{\mathrm{d}t}+1\times(i_{\mathrm{L}}-1\times\dfrac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t})=0 \\ v_{\mathrm{C}}+1\times(1\times\dfrac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t}-i_{\mathrm{L}})+1\times(1\times\dfrac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t})=0 \end{cases}$$

$$\frac{\mathrm{d}^2v_{\mathrm{C}}}{\mathrm{d}t^2}+2\frac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t}+v_{\mathrm{C}}=0$$

$$v_{\mathrm{C}}(t)=(K_1+K_2t)\mathrm{e}^{-t}$$

The equivalent circuit at t=0⁺ is shown in Figure P-27(c),

$$\begin{cases} v_{\mathrm{C}}(0^+)=5 \\ \dfrac{\mathrm{d}v_{\mathrm{C}}(0^+)}{\mathrm{d}t}=\dfrac{i_{\mathrm{C}}(0^+)}{1}=\dfrac{-\dfrac{5}{1+1}}{1}=-2.5\mathrm{V/s} \end{cases}$$

$$\begin{cases} 5=K_1 \\ -2.5=-K_1+K_2 \end{cases}$$

$$\begin{cases} K_1=5 \\ K_2=2.5 \end{cases}$$

$$v_{\mathrm{C}}(t)=(5+2.5t)\mathrm{e}^{-t}$$

$$\begin{aligned} v_0(t)&=1\times(1\times\frac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t})+v_{\mathrm{C}} \\ &=[2.5\mathrm{e}^{-t}-(5+2.5t)\mathrm{e}^{-t}]+(5+2.5t)\mathrm{e}^{-t} \\ &=2.5\mathrm{e}^{-t}\mathrm{V},t>0 \end{aligned}$$

（b）List differential equations

$$\begin{cases} v_{\mathrm{C}}+2\times(1\times\dfrac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t}-i_{\mathrm{L}})+1\times(1\times\dfrac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t})=0 \\ 2\times\dfrac{\mathrm{d}i_{\mathrm{L}}}{\mathrm{d}t}+1\times i_{\mathrm{L}}+2\times(i_{\mathrm{L}}-1\times\dfrac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t})=0 \end{cases}$$

$$6\frac{\mathrm{d}^2i_{\mathrm{L}}}{\mathrm{d}t^2}+7\frac{\mathrm{d}i_{\mathrm{L}}}{\mathrm{d}t}+3i_{\mathrm{L}}=0$$

The characteristic equation is

$$6s^2+7s+3=0$$

$$\begin{cases} s_1=\dfrac{-7+\mathrm{j}\sqrt{23}}{12}=-0.58+\mathrm{j}0.4 \\ s_2=\dfrac{-7-\mathrm{j}\sqrt{23}}{12}=-0.58-\mathrm{j}0.4 \end{cases}$$

$$i_{\mathrm{L}}(t)=K_1\mathrm{e}^{(-0.58+\mathrm{j}0.4)t}+K_2\mathrm{e}^{(-0.58-\mathrm{j}0.4)t}$$

The equivalent circuit at t=0⁺ is shown in Figure P-27(d),

$$v_L(0^+) = -\frac{2}{1+2}\times 5 = -\frac{10}{3}$$

$$\begin{cases} i_L(0^+) = 0 \\ \dfrac{\mathrm{d}i_L(0^+)}{\mathrm{d}t} = \dfrac{(-\frac{10}{3})}{2} = -\dfrac{5}{3} \end{cases}$$

$$0 = K_1 + K_2$$

$$-\frac{5}{3} = (-0.58 + \mathrm{j}0.4)K_1 + (-0.58 - \mathrm{j}0.4)K_2$$

$$\begin{cases} K_1 = \mathrm{j}2.1 \\ K_2 = -\mathrm{j}2.1 \end{cases}$$

$$i_L(t) = \mathrm{j}2.1\mathrm{e}^{(-0.58+\mathrm{j}0.4)t} - \mathrm{j}2.1\mathrm{e}^{(-0.58-\mathrm{j}0.4)t}$$
$$= -4.2\mathrm{e}^{-0.58t}\sin 0.4t$$

$$v_0(t) = 1\times i_L(t)$$
$$= -4.2\mathrm{e}^{-0.58t}\sin 0.4t\ \mathrm{V}, t > 0$$

3.28 For the circuit in Figure P3-28，find $v_C(t)$ if $v_C(0^+)$=1V, $i_C(0^+)$=2A, L=1/4 H, C=1 F. Use the following resistor values respectively:

(1) R=1/5Ω.

(2) R=1/4Ω.

(3) R=1/2Ω

(4) $R\to\infty$

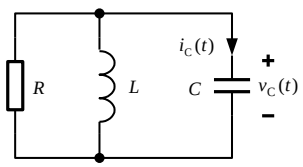


Figure P3-28

Solution

List differential equation for v_C

$$C\frac{\mathrm{d}v_C}{\mathrm{d}t} + \frac{v_C}{R} + i_L(0^+) + \frac{1}{L}\int_{0^+}^{t} v_C(\tau)\mathrm{d}\tau = 0$$

$$\frac{\mathrm{d}^2 v_C}{\mathrm{d}t^2} + \frac{1}{RC}\frac{\mathrm{d}v_C}{\mathrm{d}t} + \frac{1}{LC}v_C = 0$$

（1）R=1/5Ω.

$$\frac{\mathrm{d}^2 v_C}{\mathrm{d}t^2} + 5\frac{\mathrm{d}v_C}{\mathrm{d}t} + 4v_C = 0$$

The characteristic equation is

$$s^2+5s+4=0$$

$$\begin{cases} s_1=-1 \\ s_2=-4 \end{cases}$$

$$v_C(t)=K_1e^{-t}+K_2e^{-4t}$$

$$\begin{cases} 1=K_1+K_2 \\ \frac{2}{1}=-K_1-4K_2 \end{cases}$$

$$\begin{cases} K_1=2 \\ K_2=-1 \end{cases}$$

$$v_C(t)=2e^{-t}-e^{-4t}\text{V},t>0$$

（2）R=1/4Ω.

$$\frac{d^2v_C}{dt^2}+4\frac{dv_C}{dt}+4v_C=0$$

$$s^2+4s+4=0$$

$$\begin{cases} s_1=-2 \\ s_2=-2 \end{cases}$$

$$v_C(t)=K_1e^{-2t}+K_2te^{-2t}$$

$$\begin{cases} 1=K_1 \\ \frac{2}{1}=-2K_1+K_2 \end{cases}$$

$$\begin{cases} K_1=1 \\ K_2=4 \end{cases}$$

$$v_C(t)=(1+4t)e^{-2t}\text{V},t>0$$

（3）R=1/2Ω.

$$\frac{d^2v_C}{dt^2}+2\frac{dv_C}{dt}+4v_C=0$$

$$s^2+2s+4=0$$

$$\begin{cases} s_1=-1+j\sqrt{3} \\ s_2=-1-j\sqrt{3} \end{cases}$$

$$v_C(t)=K_1e^{(-1+j\sqrt{3})t}+K_2e^{(-1-j\sqrt{3})t}$$

$$\begin{cases} 1 = K_1 + K_2 \\ \dfrac{2}{1} = (-1 + \mathrm{j}\sqrt{3})K_1 + (-1 - \mathrm{j}\sqrt{3})K_2 \end{cases}$$

$$\begin{cases} K_1 = \dfrac{1}{2} - \mathrm{j}\dfrac{\sqrt{3}}{2} \\ K_2 = \dfrac{1}{2} + \mathrm{j}\dfrac{\sqrt{3}}{2} \end{cases}$$

$$\begin{aligned} v_{\mathrm{C}}(t) &= (\frac{1}{2} - \mathrm{j}\frac{\sqrt{3}}{2})\mathrm{e}^{(-1+\mathrm{j}\sqrt{3})t} + (\frac{1}{2} + \mathrm{j}\frac{\sqrt{3}}{2})e^{(-1-\mathrm{j}\sqrt{3})t} \\ &= 2\mathrm{e}^{-t}\cos(\sqrt{3}t - 60^\circ)\mathrm{V}, t > 0 \end{aligned}$$

（4）$R \to \infty$

$$\frac{\mathrm{d}^2 v_{\mathrm{C}}}{\mathrm{d}t^2} + 4v_{\mathrm{C}} = 0$$

$$s^2 + 4 = 0$$

$$\begin{cases} s_1 = -\mathrm{j}2 \\ s_2 = -\mathrm{j}2 \end{cases}$$

$$v_{\mathrm{C}}(t) = K_1\mathrm{e}^{\mathrm{j}2t} + K_2\mathrm{e}^{-\mathrm{j}2t}$$

$$\begin{cases} 1 = K_1 + K_2 \\ \dfrac{2}{1} = \mathrm{j}2K_1 - \mathrm{j}2K_2 \end{cases}$$

$$\begin{cases} K_1 = \dfrac{1}{2} - \mathrm{j}\dfrac{1}{2} \\ K_2 = \dfrac{1}{2} + \mathrm{j}\dfrac{1}{2} \end{cases}$$

$$\begin{aligned} v_{\mathrm{C}}(t) &= (\frac{1}{2} - \mathrm{j}\frac{1}{2})e^{\mathrm{j}2t} + (\frac{1}{2} + \mathrm{j}\frac{1}{2})e^{-\mathrm{j}2t} \\ &= \sqrt{2}\cos(2t - 45^\circ)\mathrm{V}, t > 0 \end{aligned}$$

3.29 For a source-free parallel RLC circuit,

(1) if R=2000Ω, L= 250mH and C=10nF, find natural frequencies and indicate the nature of response for the circuit (overdamped, critically damped, or underdamped).

(2) find R that results in a damped natural frequency of ω_{d}=12krad/s.

(3) find R that results in a critically damped response.

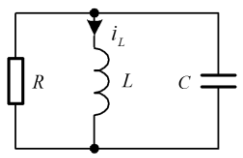


Figure P3-29

Solution

Draw a circuit for reference as shown in Figure P3-29.

(1) List differential equation for i_L

$$\frac{L\frac{di_L}{dt}}{R}+i_L+C\frac{d(L\frac{di_L}{dt})}{dt}=0$$

$$\frac{d^2i_L}{dt^2}+\frac{1}{RC}\frac{di_L}{dt}+\frac{1}{LC}i_L=0$$

$$\frac{d^2i_L}{dt^2}+5\times10^4\frac{di_L}{dt}+4\times10^8 i_L=0$$

$$s^2+5\times10^4 s+4\times10^8=0$$

$$\begin{cases}s_1=-10^4\\ s_2=-4\times10^4\end{cases}$$

It is over damped

（2）The characteristic equation is

$$s^2+\frac{1}{RC}s+\frac{1}{LC}=0$$

$$s_{1,2}=\frac{-\frac{1}{RC}\pm\sqrt{(\frac{1}{RC})^2-\frac{4}{LC}}}{2}$$

$$=-\frac{1}{2RC}\pm\sqrt{(\frac{1}{2RC})^2-\frac{1}{LC}}$$

$$\sqrt{\frac{1}{LC}-(\frac{1}{2RC})^2}=\omega$$

ω=12krad/s，

So $R=\dfrac{1}{2C\sqrt{\frac{1}{LC}-\omega^2}}=3215\,\Omega$

$$s^2+\frac{1}{RC}s+\frac{1}{LC}=0$$

$$s_{1,2}=-\frac{1}{2RC}\pm\sqrt{(\frac{1}{2RC})^2-\frac{1}{LC}}$$

$$=-16\times10^3\pm j12\times10^3$$

（3）While the circuit is critically damped，get

$$(\frac{1}{2RC})^2-\frac{1}{LC}=0$$

$$R=\frac{1}{2}\sqrt{\frac{L}{C}}=\frac{1}{2}\sqrt{\frac{250\times10^{-3}}{10\times10^{-9}}}=2.5\times10^3\ \Omega$$

3.30 The circuit in Figure P3-30 has critically damped response at t > 0. Given $v_C(0^+)$=15V,

$i_L(0^+)$=6mA,

(1) determine R.

(2) find the initial values of $i(t)$ and di/dt.

(3) find $v_C(t)$ at $t > 0$.

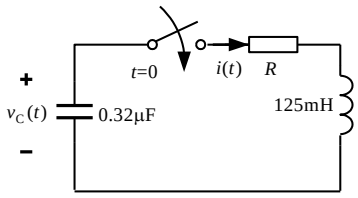


Figure P3-30

Solution

（1）List differential equation for v_C

$$L\frac{\mathrm{d}(C\frac{\mathrm{d}v_C}{\mathrm{d}t})}{\mathrm{d}t}+v_C+R\times(C\frac{\mathrm{d}v_C}{\mathrm{d}t})=0$$

$$\frac{\mathrm{d}^2v_C}{\mathrm{d}t^2}+\frac{R}{L}\frac{\mathrm{d}v_C}{\mathrm{d}t}+\frac{1}{LC}v_C=0$$

$$R=2\sqrt{\frac{L}{C}}=2\sqrt{\frac{125\times10^{-3}}{0.32\times10^{-6}}}=1250\ \Omega$$

（2） $i(0^+)=6\mathrm{mA}$

$$\frac{\mathrm{d}i(0^+)}{\mathrm{d}t}=\frac{v_L(0^+)}{L}=\frac{15-1250\times6\times10^{-3}}{125\times10^{-3}}=60\,\mathrm{A/s}$$

（3） $v_C(0^+)=15\mathrm{V}$

$$v_C'(0^+)=\frac{-i(0^+)}{C}=\frac{-6\times10^{-3}}{0.32\times10^{-6}}=-18750\ \mathrm{V/s}$$

$$\frac{\mathrm{d}^2v_C}{\mathrm{d}t^2}+10^4\frac{\mathrm{d}v_C}{\mathrm{d}t}+25\times10^6=0$$

$$s_{1,2}=-5000$$

$$v_C(t)=(K_1+K_2t)\mathrm{e}^{-5000t}$$

$$\begin{cases}K_1=15\\-5000\,K_1+K_2=-18750\end{cases}\Rightarrow\begin{cases}K_1=15\\K_2=56250\end{cases}$$

$$v_C(t)=(15+56250\,t)\mathrm{e}^{-5000t}\mathrm{V},\quad t>0$$

3.31 Find zero-state response of $v_C(t)$ at $t > 0$ in Figure P3-31.

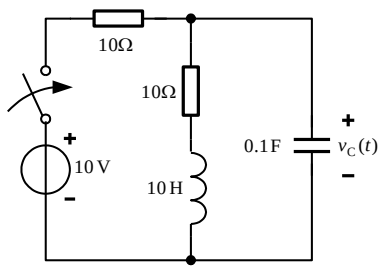
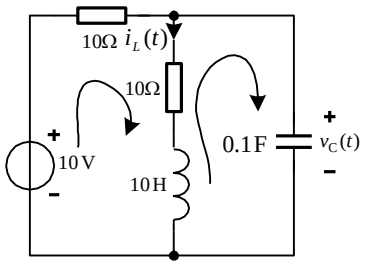


Figure P3-31



(b)

Solution

According to Figure P3-31(b), list differential equation

$$\begin{cases} 10\dfrac{\mathrm{d}i_L}{\mathrm{d}t} + 10i_\mathrm{L} + 10\times(i_\mathrm{L} + 0.1\dfrac{\mathrm{d}v_\mathrm{C}}{\mathrm{d}t}) = 10 \\ v_\mathrm{C} - 10\dfrac{\mathrm{d}i_\mathrm{L}}{\mathrm{d}t} - 10i_\mathrm{L} = 0 \end{cases}$$

$$\frac{\mathrm{d}^2 v_\mathrm{C}}{\mathrm{d}t^2} + 2\frac{\mathrm{d}v_\mathrm{C}}{\mathrm{d}t} + 2v_\mathrm{C} = 10$$

$$s^2 + 2s + 2 = 0$$

$$\begin{cases} s_1 = \dfrac{-2+\sqrt{2^2-4\times 2}}{2} = -1+\mathrm{j} \\ s_2 = \dfrac{-2-\sqrt{2^2-4\times 2}}{2} = -1-\mathrm{j} \end{cases}$$

$$\begin{aligned} v_\mathrm{C}(t) &= K_1\mathrm{e}^{(-1+\mathrm{j})t} + K_2\mathrm{e}^{(-1-\mathrm{j})t} + A \\ &= K_1\mathrm{e}^{(-1+\mathrm{j})t} + K_2\mathrm{e}^{(-1-\mathrm{j})t} + 5 \end{aligned}$$

$$\begin{cases} v_\mathrm{C}(0^+) = 0 \\ \dfrac{\mathrm{d}v_\mathrm{C}(0^+)}{\mathrm{d}t} = \dfrac{i_\mathrm{C}(0^+)}{0.1} = \dfrac{\frac{10}{10}}{0.1} = 10\,\mathrm{V/s} \end{cases}$$

$$\begin{cases} 0 = K_1 + K_2 + 5 \\ 10 = (-1+\mathrm{j})K_1 + (-1-\mathrm{j})K_2 \end{cases}$$

$$\begin{cases} K_1 = -2.5 - \mathrm{j}2.5 \\ K_2 = -2.5 + \mathrm{j}2.5 \end{cases}$$

$$\begin{aligned} v_\mathrm{C}(t) &= (-2.5-\mathrm{j}2.5)\mathrm{e}^{(-1+\mathrm{j})t} + (-2.5+\mathrm{j}2.5)\mathrm{e}^{(-1-\mathrm{j})t} + 5 \\ &= 5\sqrt{2}e^{-t}\cos(t-135^\circ) + 5\mathrm{V} \end{aligned}$$

3.32 Find $v_0(t)$ at $t > 0$ in Figure P3-32. Assume the circuit does not have initial energy storage.

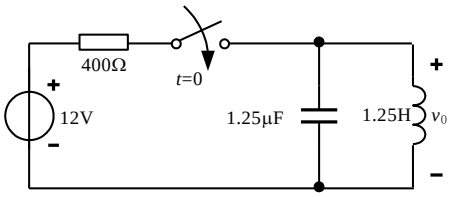
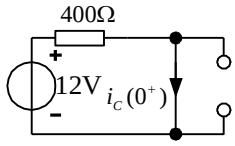


Figure P3-32



(b)

Solution

List differential equation for v_L

$$1.25\times10^{-6}\frac{\mathrm{d}v_0}{\mathrm{d}t}+i_L(0+)+\frac{1}{1.25}\int_{0^+}^{t}v_0(\tau)\mathrm{d}t+\frac{v_0-12}{400}=0$$

$$\frac{\mathrm{d}^2v_0}{\mathrm{d}t^2}+2\times10^3\frac{\mathrm{d}v_0}{\mathrm{d}t}+64\times10^4v_0=0$$

$$s^2+2\times10^3s+64\times10^4=0$$

$$\begin{cases}s_1=-400\\ s_2=-1600\end{cases}$$

$$v_0(t)=K_1\mathrm{e}^{-400t}+K_2\mathrm{e}^{-1600t}$$

The equivalent circuit at t=0⁺is shown is Figure P3-32(b),

$$\begin{cases}v_0(0^+)=0\\ \dfrac{\mathrm{d}v_0(0^+)}{\mathrm{d}t}=\dfrac{\frac{12}{400}}{1.25\times10^{-6}}=24\times10^3\end{cases}$$

Apply to complete response, get

$$\begin{cases}0=K_1+K_2\\ 24000=-400K_1-1600K_2\end{cases}$$

$$\begin{cases}K_1=20\\ K_2=-20\end{cases}$$

$$v_0(t)=20(\mathrm{e}^{-400t}-\mathrm{e}^{-1600t})\mathrm{V},t>0$$

3.33 The circuit in Figure P3-33 has been in steady state and suddenly the source voltage changes to 250V at t = 0. Find $v_C(t)$ at t > 0.

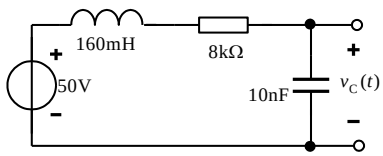


Figure P3-33

Solution

The differential equation when voltage source changes to 250V is

$$160\times10^{-3}\frac{\mathrm{d}(10\times10^{-9}\frac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t})}{\mathrm{d}t}+8\times10^{3}\times(10\times10^{-9}\frac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t})+v_{\mathrm{C}}=250$$

$$\frac{\mathrm{d}^2v_C}{\mathrm{d}t^2}+5\times10^4\frac{\mathrm{d}v_{\mathrm{C}}}{\mathrm{d}t}+6.25\times10^8v_{\mathrm{C}}=15.625\times10^{10}$$

$$s^2+5\times10^4s^2+6.25\times10^8=0$$

$$s_{1,2}=\frac{-5\times10^4\pm\sqrt{(5\times10^4)^2-4\times6.25\times10^8}}{2}=-2.5\times10^4$$

$$\begin{aligned}v_{\mathrm{C}}(t)&=(K_1+K_2t)\mathrm{e}^{-2.5\times10^4t}+A\\&=(K_1+K_2t)\mathrm{e}^{-2.5\times10^4t}+250\end{aligned}$$

From initial value, get

$$\begin{cases}v_{\mathrm{C}}(0^+)=50\\\dfrac{\mathrm{d}v_{\mathrm{C}}(0^+)}{\mathrm{d}t}=0\end{cases}$$

Apply to complete response, get

$$\begin{cases}50=K_1+250\\0=-25000\ K_1+K_2\end{cases}$$

$$\begin{cases}K_1=-200\\K_2=-5\times10^6\end{cases}$$

$$v_{\mathrm{C}}(t)=250-(200+5\times10^6t)\mathrm{e}^{-2.5\times10^4t}\mathrm{V},t>0$$